CAPSTONE 04 · VIBE-BUILT TOOL · SOLUTION BRIEF

How to build, review and share a small tool, step by step.

Nine steps. You do not need to write code, but you do need to say clearly what you want, use what comes back, and check it before anyone else sees it.

TOTAL TIME	STEPS	WORKED EXAMPLE	PLAN
About five hours	Nine, in order	Meera's ad budget splitter	Pro, Max, Team or Enterprise

THE LOOP YOU WILL RUN

STEP 01	STEPS 02-03	STEPS 04-05	STEP 06	STEPS 07-09
Spec	**Build v1**	**Use and fix**	**Review**	**Share and note**
One paragraph: who, the one job, inputs, outputs, done.	Claude Code plans first, then builds one HTML file.	Use it on real numbers. Fix three things. Twice.	Check it the way a stranger would. You own it.	One outside tester, one change, one page.

THE WEEK

DAY 1	DAY 2	DAY 3	DAY 4	DAY 5
Spec and set up	**Version 2**	**Version 3**	**Review and share**	**Note**
Write the spec, install Claude Code, build v1.	Use it for real, fix three things.	Edge cases, phone, look and feel.	Run the review, send it to a tester.	Make the tester's change, write it up.
Steps 01-03 · 1.5 h	Step 04 · 45 min	Step 05 · 45 min	Steps 06-08 · 1 h	Step 09 · 1 h

BEFORE YOU START

- [ ] A Pro, Max, Team or Enterprise plan to log in to Claude Code.
- [ ] A computer where you can install software, or the Claude desktop app.
- [ ] One real job you repeat with a sheet or a calculator today.
- [ ] Real numbers or content to test it with.
- [ ] One person outside the course willing to try it for ten minutes.

WORKED EXAMPLE · MEERA

Each month Meera splits Lumen's ad budget across Meta, Google and marketplace ads in a spreadsheet that only she understands. She builds a one-page budget splitter she can use herself and hand to her team.

Menus and commands change between releases. If one differs, check the Claude Code docs.

01

20 MINUTES · DAY 1

Write a one-paragraph spec.

The spec shapes version one more than anything else. One screen, one job; add more once it works.

TEMPLATE · THE SPEC

A [tracker / calculator / dashboard / page] for [who] that [the one job].
They enter: [inputs]. It shows: [outputs]. The rule it follows: [the calculation or logic, in plain words].
Done means: [a worked example with the right answer].
Constraints: one HTML file, no login, no outside services, works on a phone.

WORKED EXAMPLE · MEERA'S SPEC

"A calculator for Lumen's marketing team that splits next month's ad budget across Meta, Google and marketplace ads. They enter the total budget and, for each channel, last month's spend and revenue. It shows each channel's return on ad spend, the suggested split in rupees and per cent, and the revenue expected if returns hold. The rule: split in proportion to return on ad spend, but no channel below 10% or above 60%. Done means: with ₹5,00,000 and last month's figures, it matches the split in my spreadsheet to the rupee. One HTML file, no login, no outside services, works on a phone."

× AVOID Skipping the worked example. Without a known right answer, you cannot tell if the tool is correct.

✓ CHECK Your spec fits in one paragraph and includes one worked example with the correct answer.

02

20 MINUTES · DAY 1 · MODULE 5.3

Set up Claude Code in an empty folder.

Prefer no terminal? The Claude desktop app runs Claude Code too. Otherwise, install it:

TERMINAL

```
# macOS, Linux, WSL
curl -fsSL https://claude.ai/install.sh | bash
# Windows PowerShell
irm https://claude.ai/install.ps1 | iex

# open a NEW terminal window, then check it installed
claude --version

# make a folder for the tool and start Claude Code inside it
mkdir budget-splitter
cd budget-splitter
claude
```

01 On the first run it asks you to log in with your Claude account; `/login` switches later. It edits files in the folder you start it in, so always start in the tool's own folder.

✓ CHECK **`claude --version`** prints a version number, and Claude Code opens in your empty folder.

03

45 MINUTES · DAY 1 · MODULE 5.3

Build version one, plan first.

Ask for a plan before any code. Reading a ten-line plan is the cheapest place to catch a misunderstanding.

PROMPT · FIRST MESSAGE

Here is the spec for a small tool: [paste spec]
Before writing any code, give me a short plan: what is on the screen, the inputs, the calculation step by step, and how I can check it works using my worked example. Wait for my OK.

01 Read the plan. Check the calculation against your own understanding, in plain words. Correct anything wrong.

02 Approve: *"Go ahead. Build it as a single index.html with no outside libraries and no network calls. Then save a copy as v1.html."*

03 Open **`index.html`** in your browser by double-clicking it in the folder. Enter your worked example.

✓ CHECK v1.html exists, opens in a browser, and you have written down whether it gets your worked example right.

04

45 MINUTES · DAY 2

Use it for real, then fix three things.

Use version one on real work, not on a demo. Write down every point where it is wrong, confusing or slow. Pick the three that matter most and ask for exactly those.

PROMPT · EACH ITERATION

I used v[n]. Change only these three things: 1. [problem, and what should happen] 2. […] 3. […]
Do not change anything else. When done, save a copy as v[n+1].html and list what you changed.

WORKED EXAMPLE · WHAT MEERA FOUND IN V1, AND ASKED FOR

PROBLEM IN V1	WHAT SHE ASKED FOR IN V2
60% cap ignored: Meta got 74%	Apply the 10% floor and 60% cap, then share the rest in proportion
Rupees showed 8 decimals	Whole rupees, with Indian digit grouping: ₹5,00,000
A blank field showed "NaN"	Show "Enter last month's spend" under the empty field

× AVOID "Make it better." Vague requests produce changes you did not ask for and cannot trace.

✓ CHECK v2.html exists, and each of your three problems is fixed when you retest it.

05

45 MINUTES · DAY 3

Version three: edge cases, phone, finish.

Version two works for you. Version three works for someone who is not you. Try the inputs a stranger would, then ask for the fixes in one iteration prompt.

INPUTS TO TRY

- Every field empty
- Zero, a negative number, and a very large number
- Text where a number should be
- Commas and the ₹ sign typed in
- A channel with spend but no revenue

FINISH TO CHECK

- Readable on a phone: open the file on yours
- Every input has a visible label
- One line explaining how it works
- Your brand's colours and font, if you have them
- A way to take the result away: copy or print

WORKED EXAMPLE · MEERA'S V3 REQUEST

"Change only: 1. A channel with spend and no revenue crashes the split. Treat its return as zero, so it gets the 10% floor. 2. On a phone the table runs off the screen. Stack the channels instead. 3. Add a 'Copy as table' button so I can paste the plan into our sheet, and one line under the title explaining the rule. Use colours #191817, #FAFAF7 and #BD5A38. Save as v3.html."

✓ CHECK v3.html handles every input on the left without breaking, and passes every finish check on the right.

06

30 MINUTES · DAY 4

Review it as if your name is on it.

It is. You do not need to read the code, but you do need evidence that it is right and safe. Use Claude Code to explain, and check the answers yourself.

CHECK	HOW
The maths is right	Work out two cases by hand or in a sheet. Ask: "Show the calculation for these inputs, step by step." Compare.
Nothing leaves the page	Ask: "List every outside file this page loads and every place it sends data." The answer should be none.
No secrets inside	Search the file for "key", "token", "password" and any real names or email addresses.
Every control works	Click every button and change every field, on a laptop and a phone.
You can explain it	Ask: "Explain how this file works in five plain sentences." If you cannot follow it, ask again.

✓ CHECK All five checks pass, and you have noted what you found. Fix anything that failed before you share it.

07

15 MINUTES · DAY 4

Export it, or turn it into a link.

ROUTE A · THE FILE

Send the HTML file

01 Your tool is already a single file: `index.html`, the same as your final version.

02 Send it as an attachment. It opens in any browser by double-clicking, with no install.

ROUTE B · A LINK

Make it an artifact

01 Open a chat in Claude, attach `index.html`, and ask: *"Show this file as an artifact, unchanged."*

02 Check it works there, then use the artifact's share or publish option to get a link.

✓ CHECK Open what you are about to send, on a device you did not build it on. It works there.

08

30 MINUTES · DAY 4 · TESTER'S TIME: 10 MINUTES

Put it in front of one outside person.

01 Choose someone outside the course who would plausibly use a tool like this. A colleague in the right role is better than a friend being kind.

02 Give them the task, not a tour: *"You have ₹5,00,000 for next month's ads. Use this to decide the split."* Then stay quiet.

03 Note where they hesitate, misread or click the wrong thing. That is the most useful data you will get.

04 Afterwards, ask the four questions below, and write their answers down in their words.

05 Make the one change their reaction most clearly calls for, and save it as the final version.

THE FOUR QUESTIONS

- What did you think this does, before you used it?
- Where did you get stuck or unsure?
- What would you change first?
- Would you use it? What would make you?

WORKED EXAMPLE · MEERA'S TESTER

Kabir, who runs marketing at another brand, used it in six minutes. In his words: "Useful, but I didn't realise I could change the 10% and 60% limits." Meera made the limits editable fields with labels, and saved v4.

× AVOID Explaining as they go. Every explanation hides a problem a real user would hit alone.

✓ CHECK Your tester's answers are written down in their words, and one change is made because of them.

09

45 MINUTES · DAY 5

Write the note, then submit.

One page. Lead with what the tool does, then show how it changed. The iteration table is the heart of it.

SECTION	WHAT GOES IN IT
1 · The tool	What it does, who it is for, and the job it replaces
2 · Iterations	A table: version, what was wrong, what you asked for, and why
3 · Review	The five checks from step 06, and anything you found and fixed
4 · Tester	Who they were, what they said in their words, and the change you made
5 · Next	What you would build next if you kept going

WORKED EXAMPLE · MEERA'S ITERATION TABLE

VERSION	WHAT WAS WRONG	WHAT CHANGED
v1 → v2	Cap ignored, raw decimals, "NaN" on blanks	Floor and cap applied, rupee formatting, field messages
v2 → v3	Crashed on zero revenue, unusable on a phone	Zero handled, stacked phone layout, copy button, brand colours
v3 → v4	Tester did not see the limits could change	Limits made labelled, editable fields

BEFORE YOU SUBMIT

- [ ] Final HTML file, or a working link
- [ ] v1, v2 and v3 saved, each with a reason
- [ ] Worked example gives the right answer
- [ ] All five review checks passed
- [ ] Outside tester's words, and your change
- [ ] No keys, passwords or personal data in the file

IF SOMETHING GOES WRONG

"claude: command not found"	Open a new terminal window. If it persists, run the installer again.
Login fails	Type /login inside Claude Code and follow the browser prompt.
The page is blank	Tell Claude Code exactly what you see and ask it to find the cause.
It changed things you did not ask	Say "change only" and list the items. Restore from your saved version.
The numbers are wrong	Give it your worked example with the correct answer, and ask why they differ.